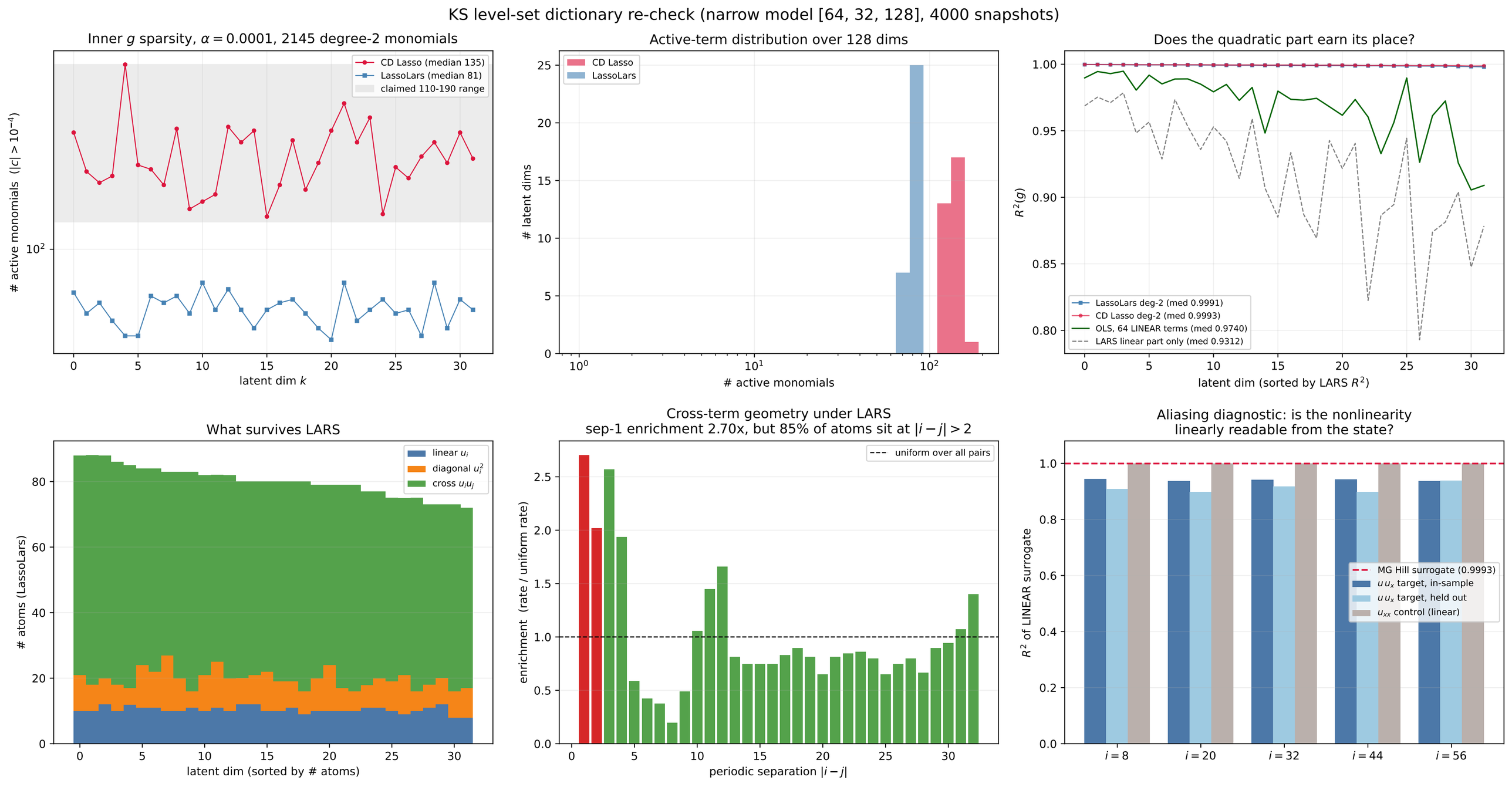
KS level-set dictionary re-check (narrow model [64, 32, 128], 4000 snapshots)
Inner g sparsity, α = 0.0001, 2145 degree-2 monomials
CD Lasso (median 135)
LassoLars (median 81)
claimed 110-190 range
active monomials (|c| > 10^-4)
latent dim k
Active-term distribution over 128 dims
CD Lasso
LassoLars
latent dims
active monomials
Does the quadratic part earn its place?
R²(g)
LassoLars deg-2 (med 0.9991)
CD Lasso deg-2 (med 0.9993)
OLS, 64 LINEAR terms (med 0.9740)
LARS linear part only (med 0.9312)
latent dim (sorted by LARS R²)
What survives LARS
linear u_i
diagonal u_i²
cross u_i u_j
atoms (LassoLars)
latent dim (sorted by # atoms)
Cross-term geometry under LARS
sep-1 enrichment 2.70x, but 85% of atoms sit at |i − j| > 2
uniform over all pairs
enrichment (rate / uniform rate)
periodic separation |i − j|
Aliasing diagnostic: is the nonlinearity
linearly readable from the state?
R² of LINEAR surrogate
MG Hill surrogate (0.9993)
u u_x target, in-sample
u u_x target, held out
u_xx control (linear)
i = 8
i = 20
i = 32
i = 44
i = 56